

Tchebychev's Inequality

A Fundamental Tool in Mathematical Analysis

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What is a σ -Algebra?

Definition (σ -Algebra)

Let Ω be a non-empty set. A collection $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is called a **σ -algebra** if:

- 1 $\Omega \in \mathcal{F}$ (contains the whole space)
- 2 If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (closed under complements)
- 3 If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ (closed under countable unions)

Examples

- $\mathcal{F} = \{\emptyset, \Omega\}$ — the **trivial** σ -algebra.
- $\mathcal{F} = \mathcal{P}(\Omega)$ — the **power set** (all subsets).
- $\mathcal{F} = \mathcal{B}(\mathbb{R})$ — the **Borel σ -algebra** (generated by open sets in $\mathbb{R}$).

What is a Measure?

Definition (Measure)

Let $(\Omega, \mathcal{F})$ be a measurable space. A function $\mu : \mathcal{F} \rightarrow [0, \infty]$ is a **measure** if:

- ① $\mu(\emptyset) = 0$ (empty set has zero measure)
- ② **Countable additivity:** If $A_1, A_2, \dots \in \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

Examples

- **Lebesgue measure** on $\mathbb{R}$: $\mu([a, b]) = b - a$ (length).
- **Counting measure:** $\mu(A) = |A|$ (number of elements).
- **Dirac measure:** $\delta_x(A) = 1$ if $x \in A$, else 0.

The Measure Space $(\Omega, \mathcal{F}, \mu)$

Definition (Measure Space)

A **measure space** is a triple $(\Omega, \mathcal{F}, \mu)$ where:

- Ω — a non-empty set (the “universe”)
- $\mathcal{F}$ — a σ -algebra on Ω (the “measurable sets”)
- μ — a measure on $\mathcal{F}$ (assigns “size” to sets)

Special Cases

- If $\mu(\Omega) = 1$: called a **probability space**.
- If $\mu(\Omega) < \infty$: called a **finite measure space**.
- $(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$ where m is Lebesgue measure: the standard setting for real analysis.

Measurable Functions

Definition (Measurable Function)

A function $f : \Omega \rightarrow \mathbb{R}$ is $\mathcal{F}$ -**measurable** if for every $\alpha \in \mathbb{R}$:

$$\{f > \alpha\} := \{\omega \in \Omega : f(\omega) > \alpha\} \in \mathcal{F}.$$

Why this matters: Measurability ensures that “level sets” like $\{|f| \geq \lambda\}$ belong to $\mathcal{F}$, so we can apply the measure μ to them.

Examples

- Continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are always Borel measurable.
- Monotone functions on $\mathbb{R}$ are Borel measurable.
- Limits of measurable functions are measurable.

The Lebesgue Integral

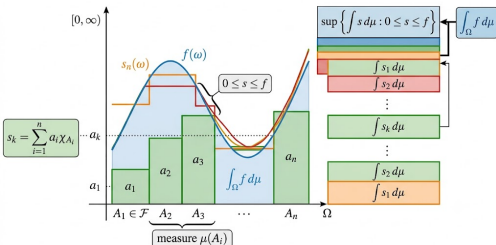
Definition (Lebesgue Integral)

For $f \geq 0$ measurable:

$$\int_{\Omega} f d\mu = \sup \left\{ \int_{\Omega} s d\mu : 0 \leq s \leq f \right\}$$

where $s = \sum_{i=1}^n a_i \chi_{A_i}$ is simple.

The Lebesgue Integral as the Supremum of Under-approximating Simple Functions



Monotonicity

$$0 \leq f \leq g \implies \int f d\mu \leq \int g d\mu.$$

Key Fact

Monotonicity is the **engine** behind Markov's and Tchebychev's inequalities.

Definition (L^p Space)

For $1 \leq p < \infty$, the **Lebesgue space** $L^p(\Omega, \mu)$ is:

$$L^p(\Omega, \mu) = \left\{ f \text{ measurable} : \int_{\Omega} |f|^p d\mu < \infty \right\}$$

equipped with the L^p **norm**:

$$\|f\|_p = \left(\int_{\Omega} |f|^p d\mu \right)^{1/p}.$$

Intuition

- $f \in L^1$: the function is “absolutely integrable.”
- $f \in L^2$: the function has “finite energy” — this is the natural setting for Tchebychev.

Markov's Inequality — Statement

The starting point for Tchebychev is this simpler result:

Theorem (Markov's Inequality)

Let $f : \Omega \rightarrow [0, \infty]$ be a non-negative measurable function on $(\Omega, \mathcal{F}, \mu)$. For any $\alpha > 0$:

$$\mu(\{f \geq \alpha\}) \leq \frac{1}{\alpha} \int_{\Omega} f \, d\mu$$

In Words

The “size” of the set where f is large is controlled by the integral of f .

Large values cannot occur on a large set if the integral is finite.

Markov's Inequality — Proof

Proof.

Let $A = \{f \geq \alpha\} = \{\omega \in \Omega : f(\omega) \geq \alpha\}$.

Step 1: Since $f \geq 0$, restricting the domain of integration only decreases the integral:

$$\int_{\Omega} f \, d\mu \geq \int_A f \, d\mu.$$

Step 2: On the set A , we know $f(\omega) \geq \alpha$, so:

$$\int_A f \, d\mu \geq \int_A \alpha \, d\mu = \alpha \cdot \mu(A).$$

Step 3: Combining:

$$\int_{\Omega} f \, d\mu \geq \alpha \cdot \mu(A) \implies \mu(A) \leq \frac{1}{\alpha} \int_{\Omega} f \, d\mu.$$



Tchebychev's Inequality — Statement

Theorem (Tchebychev's Inequality)

Let $(\Omega, \mathcal{F}, \mu)$ be a finite measure space and $f \in L^2(\Omega, \mu)$.

Define the mean: $\bar{f} = \frac{1}{\mu(\Omega)} \int_{\Omega} f \, d\mu$.

Then for any $\lambda > 0$:

$$\mu(\{\omega : |f(\omega) - \bar{f}| \geq \lambda\}) \leq \frac{1}{\lambda^2} \int_{\Omega} |f - \bar{f}|^2 \, d\mu$$

In Words

The measure of the set where f deviates from its mean by at least λ is bounded by the **variance** divided by λ^2 .

Tchebychev's Inequality — Proof

Proof.

Step 1: Define $g = |f - \bar{f}|^2$. Note that $g \geq 0$ and g is measurable.

Step 2: Observe the equivalence of level sets:

$$|f - \bar{f}| \geq \lambda \iff |f - \bar{f}|^2 \geq \lambda^2 \iff g \geq \lambda^2.$$

Step 3: Apply Markov's inequality to g with threshold λ^2 :

$$\mu(\{g \geq \lambda^2\}) \leq \frac{1}{\lambda^2} \int_{\Omega} g \, d\mu.$$

Step 4: Substitute back:

$$\mu(\{|f - \bar{f}| \geq \lambda\}) \leq \frac{1}{\lambda^2} \int_{\Omega} |f - \bar{f}|^2 \, d\mu = \frac{\|f - \bar{f}\|_2^2}{\lambda^2}.$$

Why Does the Proof Work?

The proof relies on exactly **two** analytic facts:

1. Monotonicity of the Lebesgue Integral

If $f \leq g$ a.e., then $\int f \, d\mu \leq \int g \, d\mu$.

This is what makes Markov's inequality true.

2. Measurability of Level Sets

If f is measurable, then $\{|f| \geq \lambda\} \in \mathcal{F}$ for every $\lambda > 0$.

This ensures $\mu(\{|f| \geq \lambda\})$ is well-defined.

No assumptions on the structure of Ω , the form of μ , or the shape of f are needed.

Statistical Form: $k\sigma$ Version

In statistics, Tchebychev is often stated using k standard deviations:

$k\sigma$ Form

Let X have mean μ and standard deviation $\sigma > 0$. For any $k > 0$:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Example: Exam Scores

A class has mean score $\mu = 70$ and standard deviation $\sigma = 10$.

What fraction of students can score outside $[50, 90]$?

Here $|X - 70| \geq 20 = 2\sigma$, so $k = 2$:

$$P(|X - 70| \geq 20) \leq \frac{1}{2^2} = \frac{1}{4} = 25\%.$$

At least **75%** of students score between 50 and 90.

Statistical Form: Raw Distance Version

We can also state the bound directly in terms of the deviation λ :

Raw Distance Form

Let X have mean μ and variance σ^2 . For any $\lambda > 0$:

$$P(|X - \mu| \geq \lambda) \leq \frac{\sigma^2}{\lambda^2}$$

Example: Delivery Times

A courier's delivery time has mean $\mu = 30$ min and variance $\sigma^2 = 25$ min².

What is the chance of deviating by more than 15 minutes?

Set $\lambda = 15$:

$$P(|X - 30| \geq 15) \leq \frac{25}{15^2} = \frac{25}{225} = \frac{1}{9} \approx 11.1\%.$$

At least **88.9%** of deliveries arrive between 15 and 45 minutes.

Applications of Tchebychev's Inequality

Tchebychev's inequality is applied in the following areas:

- **Convergence Theory:** L^p convergence implies convergence in measure.
- **Weak Law of Large Numbers:** Sample averages converge to the mean.
- **Approximation Theory:** Bernstein's proof of the Weierstrass theorem.
- **Fourier Analysis:** L^2 convergence of partial sums implies convergence in measure.
- **Ergodic Theory:** Convergence of ergodic averages via L^2 bounds.
- **PDE & Sobolev Spaces:** Controlling level sets via Sobolev norms.
- **Spectral Theory:** Bounding spectral measure for self-adjoint operators.

In each case: **integral estimates** $\longrightarrow$ **pointwise control**.

Sharpness of Tchebychev's Inequality

The bound is tight!

For any $\lambda > 0$, there exists a measurable function for which **equality** holds.

Extremal Example

Let $\Omega = \{-1, 0, 1\}$ with $\mu(\{-1\}) = \mu(\{1\}) = \frac{1}{2k^2}$, $\mu(\{0\}) = 1 - \frac{1}{k^2}$.

Define $f(\omega) = \omega$. Then $\bar{f} = 0$, $\sigma^2 = 1/k^2$, and:

$$\mu(\{|f| \geq 1\}) = \frac{1}{k^2} = \frac{\sigma^2}{\lambda^2}. \quad \checkmark$$

Tighter bounds need extra assumptions (sub-Gaussian, log-concavity, etc.).

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Thank You

Questions?